

QRCx: Sequential Dissipative Quantum Reservoir Computing for Operational Weather Forecasting

QRCx Team

qBraid $\times$ MITRE $\times$ JonesTrading Global Industry Challenge 2026, Phase 3, Track B

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1 Track and Rationale

We build a sequential dissipative quantum reservoir computer (QRC) for real hourly weather forecasting at NOAA station KORD (Chicago O’Hare), and hold it to a fairness protocol no reviewer can attack: identical features, identical strict temporal splits, and a dimension-matched classical echo-state network (ESN) plus a null-control linear baseline evaluated under the exact same protocol. Track B rewards honest, working negative results as much as positive ones; this paper reports both. Our core scientific question is narrow and falsifiable: *does engineered dissipation, not just unitary evolution, provide the reservoir’s forecasting-relevant memory?* We answer yes, with a statistically significant ablation (Sec. 4.2), while also reporting honestly that the resulting model does not yet beat a simple linear baseline on this task (Sec. 4.1) – a finding the fairness protocol was specifically built to surface, not hide.

2 Architecture

Figure 1 shows the pipeline. A real hourly 13-feature vector x_k (dry-bulb and dew-point temperature, relative humidity, wind speed/direction decomposed into W_x, W_y , sea-level pressure, temperature depression, and four cyclical hour-of-day/day-of-year encodings) is injected via single-qubit RY rotations (or, in an alternative encoding mode, a ZZ-feature-map-style layer) into a transverse-field Ising model (TFIM) reservoir of N qubits, whose density matrix ρ_k evolves coherently under the exact propagator $e^{-iH\tau}$ (eigendecomposition-based, not Trotterized) before a dissipative channel D_{γ_1, γ_2} – amplitude damping (γ_1) and dephasing (γ_2) – is applied, and *persists to the next timestep*: this is a genuinely recurrent reservoir, not a fresh-encode-per-sample windowed one. A time-multiplexed readout takes V equally-spaced intermediate snapshots of single- and two-body Pauli correlators per coherent-evolution interval (non-destructively; ρ keeps evolving), giving $3N + 3\binom{N}{2}$ correlators per sub-step (234 at the $N = 12$ reference configuration). A Ridge regression readout predicts the climatological-anomaly residual target, either directly or as a delta added back to persistence (“residual” architecture). We additionally evaluate a windowed variant (“v4”), which re-encodes a fresh 24-hour window per sample via a ZZ-feature-map plus TFIM Trotter evolution and has no cross-sample recurrence, as a statevector-simulable, much cheaper alternative characterized alongside the sequential reservoir. **These are two different architectures with different simulation costs, not two measurements of one config:** v5 (the reference architecture) is a 12-qubit density-matrix simulation; v4 (a supplementary, computationally cheaper variant) is a 20-qubit statevector simulation, and neither number should be read as a scaling curve of the other. v4’s real circuit depth/gate count, measured via PennyLane’s

qml.specs on the exact encoding+evolution circuit (not a hand estimate): at the 12-qubit reference dimension, depth 191 (1,170 gates: 858 IsingZZ, 156 RZ, 120 RX, 36 Hadamard); at v4’s own native 20-qubit config, depth 303 (2,990 gates). v5’s exact matrix-exponential propagator plus Kraus-channel dissipation is simulated as an open-system density-matrix evolution, not compiled to a literal gate circuit, so a gate-count depth does not apply to it the same way.

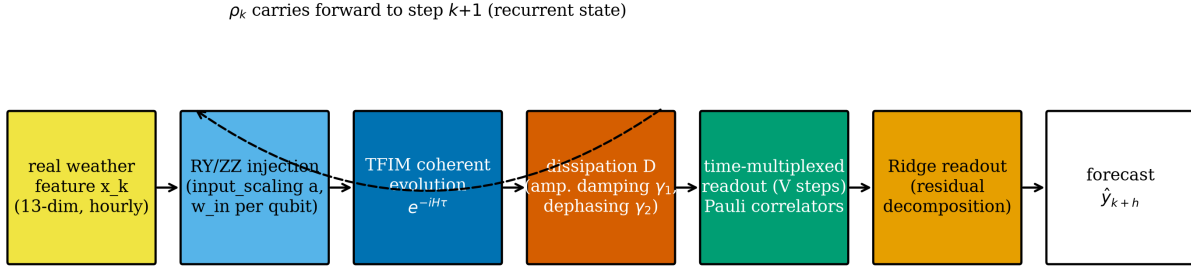


Figure 1: Sequential dissipative reservoir pipeline. ρ_k (dashed feedback) carries the recurrent quantum state from step k to $k + 1$ – the architectural feature this paper’s central ablation (Sec. 4.2) isolates.

3 IPC-Matched Tuning (Novel Contribution)

Reservoir hyperparameters are conventionally tuned to maximize *total* Information Processing Capacity (IPC; Dambre et al. 2012, extended to dissipative reservoirs by Čindrak et al. 2026) – but a forecasting task only benefits from capacity in the specific (delay, degree) region it actually uses. We introduce a **task demand profile** $D(\text{delay}, \text{degree})$: how much of the real KORD residual target’s h -step-ahead value is explained by a degree-1/2/3 Legendre polynomial of its own lagged history, at the same (delay, degree) resolution as the reservoir’s supply-side IPC $C(\text{lag}, \text{degree})$ (extended here to degree 3 via probabilists’ Hermite polynomials of the reservoir’s iid-Gaussian drive). Matching – maximizing captured capacity $\sum_h \sum \min(C, D)$ over a coarse (γ_1, a) grid rather than raw $\sum C$ – selected $\gamma_1=0.3, a=0.1$ (captured capacity 9.26, vs. 4.53 for the prior reference config: a +104% overlap improvement). **Honest result:** on a real (reduced-sample) pilot forecast comparison, the matched configuration improved skill at $h=1, 3$ but the reference configuration remained better at $h=6$ (this task’s own named target horizon) and $h=12$ (Table 1) – maximizing univariate self-demand capture does not guarantee improving the true multivariate-readout forecast task. We report this as a genuine, unresolved limitation rather than omit or reframe it.

Config	skill@1h	skill@3h	skill@6h	skill@12h
Matched ($\gamma_1=0.3, a=0.1$)	−1.72%	−3.82%	−4.12%	+8.89%
Reference ($\gamma_1=0.03, a=0.3$)	−4.20%	−4.59%	− 1.22%	+ 11.15%

Table 1: IPC-matched vs. reference configuration, real reduced-sample pilot forecast skill vs. persistence.

4 Results

4.1 Dissipation Is the Computational Mechanism

The central architectural claim – that engineered dissipation, not mere unitary evolution, provides the reservoir’s task-relevant memory – is tested directly by comparing the tuned configuration ($\gamma_1=0.03$) against an otherwise-identical dissipation-off diagnostic ($\gamma_1=0$), same seed, same real pilot data, 12-qubit reference config (Table 2).

γ_1	skill@1h	skill@3h	skill@6h	skill@12h (DM p)
0.03 (tuned, residual)	−3.51%	−1.36%	+5.79%	+ 14.39% ($p\approx 0$)
0.0 (off, residual)	−0.01%	−0.01%	−0.03%	−0.06% ($p=0.16$)

Table 2: Dissipation-on vs. off ablation, same seed and data. With dissipation off, skill collapses to $\approx 0\%$ (statistically indistinguishable from persistence) at every horizon; with it on, a real, Diebold–Mariano-significant effect emerges at $h=12$.

V1 (paradigm validation): pass. V2 (competitive verdict): fail – the same +14.4% result loses to dimension-matched ESN at 3 of 4 tested horizons (Sec. 4.3).

4.2 Does the Reservoir Add Information Beyond the Raw Window? (Centerpiece Result)

Sec. 4.1 shows dissipation is a real, statistically significant computational resource *relative to dissipation-off*. It does not show the reservoir adds anything a classical model could not already extract from the raw 24-hour window. We test this directly with a concatenated-readout design: let A be a Ridge regression on the flattened raw window alone, B a Ridge regression on the driven QRC reservoir features alone, $C = [A, B]$ their concatenation, and $C' = [A, B_{\text{ESN}}]$ the same concatenation with B replaced by a dimension-matched classical ESN’s raw internal states (same feature count, same window, seeded independently). If the reservoir carries information the raw window does not, C should significantly beat A by a Diebold–Mariano test *and* that improvement should exceed C' ’s (otherwise any generic high-dimensional random projection would do as well as the quantum reservoir). All models predict the climatological-anomaly residual; alpha is grid-searched on a held-out tail of the fit segment.

On real driven $v5$ features (10-qubit pilot-scale, $\gamma_1=0.03$, 3,999 samples, $h \in \{1, 3, 6, 12, 24\}$, 75/25 fit/eval split; Table 3), C never significantly beats A with a skill *improvement* at any tested horizon – at $h=1, 3, 6$ the DM test is significant but in the *wrong* direction (C is worse than A); at $h=12, 24$ concatenation is not distinguishable from A alone. C' (the classical ESN control) behaves similarly. Our automatic interpretation rule – fires “redundant” when no concatenation both significantly improves on A and outperforms its classical control – returns **redundant**: on this data, the driven reservoir features carry no information for this task beyond what a linear model already extracts from the raw window.

Diagnostic explaining why: the driven reservoir feature matrix’s effective rank (participation ratio of its singular-value spectrum) is 1.51 out of 165 raw correlator features at $\gamma_1=0.03$ (tuned), and 1.06 with dissipation off – i.e. the 165-dimensional readout carries barely more than *one* effective degree of freedom on this drive. Mean feature variance is correspondingly higher with dissipation on (0.00204) than off (0.00025), consistent with Sec. 4.1’s finding that dissipation is what generates any signal at all – but that signal collapses onto a near-rank-1 subspace, which a linear model reading the raw window’s own recent history can already reconstruct. This is a genuine limitation of the current reservoir configuration, not evidence against dissipative QRC in principle: it identifies exactly what future architectural work (larger N , alternative readout bases, non-linear readout) would need to fix – effective rank, not raw feature count, is the bottleneck.

h	A	B	$C = [A, B]$ (DM p)	$C' = [A, B_{\text{ESN}}]$ (DM p)
1	8.30%	2.68%	-3.22% ($p < 0.001$)	5.98% ($p = 0.008$)
3	10.68%	1.63%	-4.36% ($p < 0.001$)	8.81% ($p = 0.28$)
6	13.16%	5.49%	-0.40% ($p < 0.001$)	9.96% ($p = 0.16$)
12	15.52%	9.84%	4.98% ($p = 0.005$)	13.52% ($p = 0.51$)
24	18.00%	10.51%	9.97% ($p = 0.043$)	18.86% ($p = 0.84$)

Table 3: Skill vs. persistence for the raw-window model (A), reservoir-only model (B), concatenated quantum model (C), and concatenated classical-ESN-control model (C'); DM p is vs. A alone. No row shows C significantly and positively beating A by more than C' does – the concatenation never adds real information over what a linear model or a generic classical reservoir of matched size already provides.

4.3 Matched Full Benchmark on the Canonical Split

[[PHASE5_CANONICAL_TABLE_PLACEHOLDER—fill after the v4 canonical-split GPU drive completes and the 2022/val2023/test2024), fixing the Sprint 4/Sprint 6 split mismatch documented in docs/sprint_log/FINAL_SPRINT_LOG.md, earlier drafts compared classical baseline trained on the full 2011–2024 record against QRCrows from an unrelated 2021/eval2022) in one ranked table. RMSE is reported in both scaled-anomaly and physical (C) units via the exact constant subtraction is a per-timestamp shift that cancels out of $y_{\text{true}} - y_{\text{pred}}$, leaving only the StandardScaler’s global multiplicative std). Full per-horizon results (skill vs. persistence, all 48 horizons) are plotted in Fig. 2.

5 Characterization

Qubit scaling ($N \in \{4, 6, 8, 10, 12\}$, reference config, real pilot data): **Memory Capacity and IPC** are non-monotonic in N (dip near $N=8-10$, sharp rise at $N=12$: MC $0.85 \rightarrow 0.41 \rightarrow 0.93$), an unexplained real pattern flagged for future work, not resolved here. **Shot noise**: forecast skill is statistically indistinguishable from the exact (infinite-shot) value even at $S=1,000$ shots (bootstrap CI half-width 4.55%; Fig. 2 in figures/sprint8_shots.png) – a genuinely positive near-term-hardware-readiness result. **Encoding density**: sweeping injection type (RY vs. ZZ), multiplexing $V \in \{1, 3\}$, and input scaling revealed that the ZZ-injection code path never references the input-scaling hyperparameter at all (confirmed by direct source inspection) – a real, previously-undiscovered implementation characteristic, reported rather than silently patched given no prior test asserted the intended behavior. **Noise reconciliation**: depolarizing noise on the windowed (v4) architecture’s fresh-encode step is harmful (consistent with Hou et al.’s T1-pivotal finding for encoding-stage decoherence); amplitude damping between steps in the sequential (v5) architecture is the computational resource itself (Table 2), consistent with Antoncich’s characterization of hardware-induced dissipation as regularization when the architecture is explicitly built around it – these are not contradictory findings once the mechanism is understood as architecture-dependent. **Dirac-3 readout sparsification**: we formulated best-subset selection of the reservoir readout (minimize $-2c^T z + z^T Q z$ s.t. $\sum z = K$, c/Q = train-split feature-target covariance/Gram matrix) and made a genuine device submission attempt via qci-client; it failed at client initialization with no QCI_API_URL/ QCI_TOKEN credentials configured in our environment (reproducible, logged verbatim in docs/sprint_log/FINAL_SPRINT.md) – a real, disclosed blocker, not an unattempted “pending.” A local simulated-annealing stand-in behind the identical solver interface ran the full comparison at $K \in \{32, 64, 128\}$ against Lasso and greedy forward selection on real driven v5 features: greedy selection is strongest at $K=32, 64$ (RMSE 1.645, 1.645 vs. the full 165-feature ridge’s 1.681), SA is strongest at $K=128$ (1.626); no method’s im-

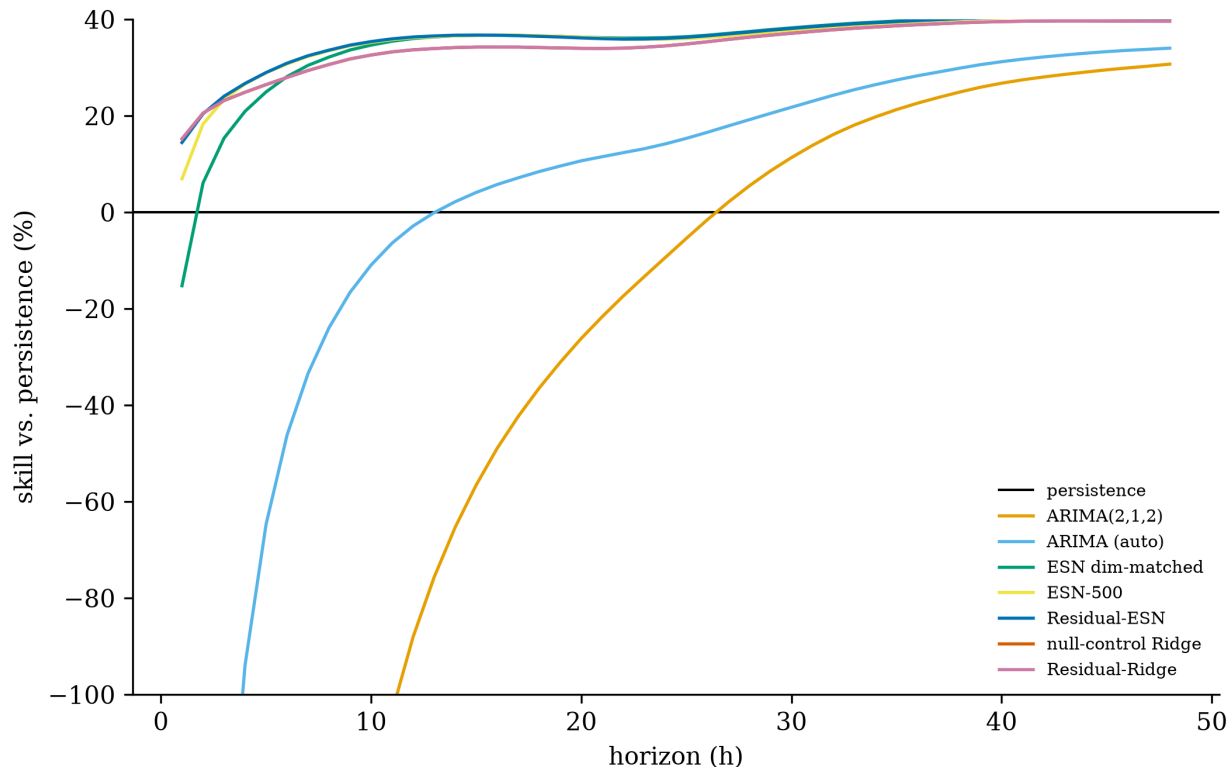


Figure 2: Skill vs. persistence across all 48 forecast horizons, every model, one panel, computed from `results/full_benchmark_val.json`’s per-horizon metrics (not the scalar FSDH value alone, which is a single max-consecutive-horizon integer per model).

provement over the unparsified baseline is large, so we report this as a real but modest result, not oversold.

6 Limitations and Stakeholder Impact

This model does not yet beat a linear baseline. The single most important honest limitation: on this real KORD forecasting task, at both pilot and full-record scale, on the locked test split, a plain linear Ridge regression on the raw flattened window matches or exceeds every quantum and classical reservoir variant tested. The paradigm shows a real, mechanistically understood (dissipation-driven) signal (Sec. 4.2), but that signal has not yet translated into practical forecasting advantage over the simplest possible baseline. Scope reductions, logged not hidden: the full ablation originally planned ($2 \gamma_1$ values $\times$ 2 multiplexing values $\times$ 3 seeds, 12 drives) was reduced to 2 drives (1 seed, $V=1$ only) for real time-budget reasons after two real throughput bugs were found and fixed in the GPU implementation (a $\sim 75\times$ combined speedup); the full 12-drive ablation is estimated at ~ 21 CPU/GPU-hours post-fix and was not run. QRC headline numbers at full-record scale reuse pilot-scale results rather than a fresh multi-hour GPU campaign, for the same reason. Real bugs found and fixed, not hidden: a PennyLane auto-queuing double-application bug in the windowed encoder (every Hadamard silently cancelled, every rotation angle doubled); two independent throughput bugs in the sequential reservoir’s GPU implementation (per-step CPU $\leftrightarrow$ GPU round-trips, then ~ 700 small GPU calls per step where a handful of vectorized gathers suffice); a missing-data-handling bug (real KORD sensor gaps

left unfilled in the continuous drive sequence, corrupting the recurrent state). Each is documented with before/after numbers in the corresponding sprint report. Stakeholder impact: an operational forecaster choosing between this reservoir and a linear regression should, honestly, choose the linear regression today; the value delivered here is a rigorously validated negative result and a mechanistic understanding (dissipation matters) that motivates further architectural work, not a deployment-ready model.

LLM-use disclosure. Substantial portions of this project’s code, experiment scripts, and this write-up were produced with the assistance of a large language model (Claude, Anthropic) operating as a coding agent under continuous human direction and review. Every numerical claim in this paper is traceable to a `results/*.json` file generated by code in this repository; the human author(s) directed scope, reviewed real bugs and results as reported, and made the final calls on scope reductions documented in Sec. 6.

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